FACULTY OF COMPUTING
UTM Johor Bahru

SCST1123: MATHEMATICS FOR SOFTWARE ENGINEERING
20242025/SEMESTER 1

ASSIGNMENT #3b (Chapter 6 & 7)

INSTRUCTION:

1. This is an assignment for a **GROUP of 3 students**. Please clearly write the course details including the group members name & matric number in the front page of the submission.
2. This assignment will give **5%** of overall course marks.
3. The submission is must be neatly written using handwriting. Any submission with handwriting that is unreadable, will be **REJECTED**.
4. Due date: **Friday - 10/01/2025, 12.00 pm**.

Question 1 **[20 Marks]**

a) Let X and Y are events in a sample space S and suppose that $P(X') = 0.30, P(Y) = 0.75,$ and $P(X' \cup Y') = 0.4$. What is $P(X \cup Y)$? (4 marks)

b) Five people have been exposed to Corona virus. Once exposed, a person has a 50-50 chance of becoming ill.

   i. What is the probability that exactly one of the people becomes ill? (3 marks)

   ii. What is the probability that none of the five people becomes ill? (2 marks)

c) A box contains 25 yellow balls and 15 white balls. Two balls are chosen at random, one after the other, without replacement.

   i. What is the probability that both balls are white? (2 marks)

   ii. What is the probability that the second ball is white, but the first ball is not? (2 marks)

   iii. What is the probability that the second ball is yellow? (3 marks)

d) A professor uses two proofreaders Ali and Hana to check a certain manuscript. Ali misses 13% of typographical errors and Hana misses 9%. Assume that the proofreaders work independently.

i. What is the probability that a randomly chosen typographical error will be missed by both proofreaders? (2 marks)

ii. If the manuscript contains 2000 typographical errors, what number can be expected to be missed? (2 marks)

Question 2 **[13 Marks]**

a) Kuala Lumpur, Johor Bahru, Penang, Kota Kinabalu, and Kuching airport among the busiest airport in Malaysia. There are direct flights connecting each city to another.

i) Draw an undirected graph model representing this flight connection. Label all edges and vertices. (4 marks)

ii) Determine the degree of each of vertex in the graph (a). (2 marks)

iii) Is the graph draw in (a-i) is a complete graph? Explain. (2 marks)

iv) Due to the COVID19 pandemic, the flight connection is reviewed. There is no direct flight to Kota Kinabalu from Penang and Johor Bahru and no direct flight to Kuching from Penang. Is the graph also as in your answer in (a-iii)? Explain. (2 marks)

b) Ahmad works at ED delivery services company. He wants to deliver 8 parcels to 8 different addresses in one housing area. Each of these houses is connected to each other via several different lanes as can be viewed in **Figure 1**.

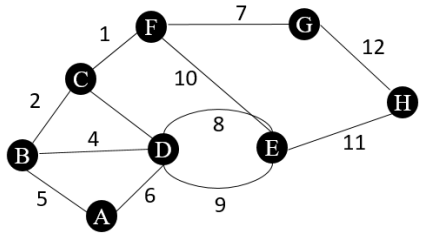


Figure 1

After completing his work, Ahmad decided to look around the area. Is it possible to tour the house area by passing all the lane (label as 1, 2, 3, 4, 5, 6, 7, 8, 9,10, 11,12) exactly once? If yes, which address he must begin and end? If not, give the reason.

(3 marks)

Question 4 **[12 Marks]**

a) The word 'WONDER' is resulted from preorder traversal of a rooted binary tree.

i. Draw the tree with height of 2 that can represent the above traversal. (1 mark)

ii. Using tree in (i) find the word resulted from the inorder traversal. (2 marks)

b) Data Communication company has offices A, B, C, D, E, F, G, H, and I. **Table 1** shows the cost, in Ringgit Malaysia (RM), of transmitting a piece of information from one of the offices to another along all existing direct links.

Table 1

From \ To	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**
A	-	4	-	-	-	-	-	8	-
B	4	-	8	-	-	-	-	11	-
C	-	8	-	7	-	4	-	-	2
D	-	-	7	-	9	14	-	-	-
E	-	-	-	9	-	10	-	-	-
F	-	-	4	14	10	-	2	-	-
G	-	-	-	-	-	2	-	1	6
H	8	11	-	-	-	-	1	-	7
I	-	-	2	-	-	-	6	7	-

i. Construct a network using vertices A to I to represent the information in **Table 1**. (3 marks)

ii. A piece of information has to be passed from office A to all the other offices, either directly or by being passed on from office to office. Use Kruskal's algorithm to find the minimum total cost of passing the information to all the offices. (6 marks)

Question 5 **[10 Marks]**

Figure 2 below shows road connecting a few locations in Johor Bahru. The numbers represent the distance between each location. Tomorrow, you want to travel from location **M** to location **S** for work purposes.

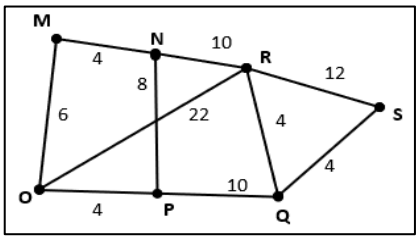


Figure 2

Find the shortest path from location **M** to **S** using Djikstra's Algorithm. (10 marks)